PAWAN AJAY PRASAD

AI Engineer | Generative AI & RAG Systems | Machine Learning

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PROFESSIONAL SUMMARY

AI Engineer with **1+ year of hands-on experience** building and deploying 3+ end-to-end **GenAI** and full-stack AI projects using **Python**, **LangChain**, **Gemini**, **ChromaDB**, and **FastAPI**. Delivered an enterprise **RAG assistant** with **100% retrieval accuracy (Hit@1/3/5)**, a **Gemini**-powered sales assistant with **Gmail/Calendar API integration**, and a **real-time collaborative workspace** with role-based permissions. Skilled in **REST APIs**, **SQL**, and testing.

SKILLS

Generative AI & AI Engineering: GenAI, RAG, LangChain, Vector Databases (ChromaDB), Semantic Search, Embeddings, LLM Integration (Gemini), Prompt Engineering, Pydantic

Backend & Full-Stack: Python, FastAPI, REST APIs, JavaScript, Node.js, WebSockets, Yjs CRDT, RBAC, Gmail & Google Calendar API Integration

Machine Learning: Scikit-learn, XGBoost, Feature Engineering, Predictive Modeling, SMOTE, Model Evaluation

Data & BI: SQL, PostgreSQL, SQLite, Pandas, NumPy, EDA, Power BI, DAX, Tableau

Testing & Tools: Pytest, GitHub, Docker, Microsoft Azure, Kubernetes, Streamlit

WORK EXPERIENCE

AlmaBetter — AI Engineering / Data Analyst Trainee

Jan 2025 – Present

- Developed AI/GenAI workflows using **Python**, **LangChain**, **Gemini**, **RAG**, embeddings, and **Streamlit** for document-based Q&A.
- Built **RAG pipelines** covering document ingestion, chunking, **semantic retrieval**, and **citation-grounded LLM responses**, improving answer accuracy for internal knowledge search.
- Applied ML, EDA, SQL, and BI across 15+ datasets leveraging **Scikit-learn/XGBoost**, **100K+ data points**, and **20+ KPIs** to surface predictive and business trends.
- Automated data-cleaning scripts across **50+ SQL queries**, reducing manual data-preparation time by roughly **30%**.
- Communicated analytical findings to a cohort of **50+ peers and mentors** to support business-ready decisions.

PROJECTS

PAWANSPACE — Collaborative Team Workspace & Document Management Platform

Nov 2026

Python, FastAPI, JavaScript, Node.js, Hocuspocus, Yjs CRDT, Tiptap, SQLite, WebSocket, Vite | GitHub

- Created a **full-stack collaborative workspace** with document management, file handling, team workspaces, and **role-based permissions**.
- Engineered a **two-service architecture** (FastAPI + Node.js Hocuspocus) separating APIs from real-time collaboration; using **Yjs CRDT**, **WebSockets**, and Tiptap for bidirectional document sync with SQLite persistence.
- Secured collaboration with **signed ticket validation** and separate application/collaboration data storage; **3 backend tests passing** and two-way in-memory CRDT synchronization verified.

AI Sales Assistant — Lead Qualification & Appointment Automation

Oct 2026

Python, Streamlit, Google Gemini, Pydantic, Gmail API, Google Calendar API, SQLite, Pytest | GitHub | Live Demo

- Designed an AI sales assistant using **Google Gemini** to handle customer conversations, extract lead information, and generate context-aware responses with **schema-validated structured outputs (Pydantic)**.
- Implemented **rule-based lead qualification** and routing (service interest, budget, timeline), separating LLM extraction from Python business logic; added **human takeover**, conversation persistence, and action logging.
- Integrated **Gmail API** (read-only inbox analysis) and **Google Calendar API** for availability checks and confirmation-gated booking with duplicate and conflict handling; **22 pytest tests passed**.

Infosys AI Knowledge Assistant — Enterprise RAG Platform

May 2026 – Aug 2026

Python, FastAPI, LangChain, Gemini, ChromaDB, RAG | GitHub | Live Demo

- Architected and delivered an enterprise RAG knowledge assistant with **LangChain**, **Gemini 3.6 Flash**, and **ChromaDB**, serving fully-cited responses across 5 department domains with **role-based access control (RBAC)**.
- Constructed PDF ingestion and vector retrieval via **PyMuPDF**, 1K/200-token chunking, **Gemini embeddings**, and department/document metadata filtering — cutting document-lookup time versus manual search.
- Achieved **100% Hit@1/3/5**, **MRR 1.0**, and access-control pass rate across **25 retrieval test cases**, with **5/5 generation evaluations** passed.

EDUCATION

M.S. in Computer Science: Machine Learning & AI Engineering, Woolf University — Jan 2025 – Present

Bachelor of Science, Mumbai University — May 2022

CERTIFICATIONS

- Microsoft Certified: Azure Fundamentals — Microsoft | View Certificate
- Advanced Certification in Data Analytics & Gen AI Engineering — NASSCOM FutureSkills Prime | View Certificate
- Applied Professional Certification in Data Science & Agentic AI — AlmaBetter | View Certificate